

Executive Report

Web3Geeks Client Inquiry Qualification Agent

1. Business Goal

Web3Geeks receives numerous client inquiries daily. Currently, sales team manually reviews each inquiry, researches the client company, and drafts responses. This process is time-consuming (15-30 minutes per inquiry), inconsistent, and cannot scale without hiring.

Solution: Automated client inquiry qualification agent that validates inquiries, researches companies, qualifies leads, drafts responses, and pauses for human approval on critical decisions.

2. Architecture

Flow: User Input -> Validate -> Research -> Qualify -> Draft -> Human Approve -> End
Error Handler catches bad input at any stage.

- Validator: Checks required fields, email format
- Researcher: Looks up company in local database
- Qualifier: Scores lead (high/medium/low)
- Drafter: Creates personalized response using LLM
- Human Review: Approves before sending

3. Framework Choice Rationale

Selected: LangGraph (with CrewAI-style role design)

- Control-heavy workflow: HITL requires precise control
- Conditional routing: Bad input -> error handler
- State management: Track data through stages
- Why NOT CrewAI: Lacks precise HITL control

4. Evaluation Results

Criterion	Score	Notes
Task Success	3/8 (38%)	Edge cases as errors (expected)
Factual Accuracy	5/5	Correct company info
Latency	8.5s avg	Within acceptable range
Cost per Run	\$0.002	Very affordable
Tone/Quality	5/5	Professional responses
Safety	100%	All tests passed safely

5. Known Limitations

- Small database: Only 5 companies in DB
- No web search: Cannot look up unknown companies
- Auto-approval: High/medium leads auto-approved
- Single LLM: No fallback if primary LLM fails
- No persistence: Results not saved to database

6. Recommended Next Steps

- Scaling: Expand DB, add web search, deploy to cloud
- Guardrails: Content filtering, rate limiting, retry logic
- Human Oversight: Email notifications, approval dashboard
- Monitoring: Dashboard, alerting, cost tracking